

OCTOBER 2021

Winning the Tech Talent Competition

Without STEM Immigration Reforms, the United States Will Not Stay ahead of China

AUTHOR

Remco Zwetsloot

A Report of the CSIS Renewing American Innovation

CSIS | CENTER FOR STRATEGIC &
INTERNATIONAL STUDIES



OCTOBER 2021

Winning the Tech Talent Competition

Without STEM Immigration Reforms, the United States Will Not Stay ahead of China

AUTHOR

Remco Zwetsloot

A Report of the CSIS Renewing American Innovation

About CSIS

The Center for Strategic and International Studies (CSIS) is a bipartisan, nonprofit policy research organization dedicated to advancing practical ideas to address the world's greatest challenges.

Thomas J. Pritzker was named chairman of the CSIS Board of Trustees in 2015, succeeding former U.S. senator Sam Nunn (D-GA). Founded in 1962, CSIS is led by John J. Hamre, who has served as president and chief executive officer since 2000.

CSIS's purpose is to define the future of national security. We are guided by a distinct set of values—nonpartisanship, independent thought, innovative thinking, cross-disciplinary scholarship, integrity and professionalism, and talent development. CSIS's values work in concert toward the goal of making real-world impact.

CSIS scholars bring their policy expertise, judgment, and robust networks to their research, analysis, and recommendations. We organize conferences, publish, lecture, and make media appearances that aim to increase the knowledge, awareness, and salience of policy issues with relevant stakeholders and the interested public.

CSIS has impact when our research helps to inform the decisionmaking of key policymakers and the thinking of key influencers. We work toward a vision of a safer and more prosperous world.

CSIS does not take specific policy positions; accordingly, all views expressed herein should be understood to be solely those of the author(s).

© 2021 by the Center for Strategic and International Studies. All rights reserved.

Acknowledgments

This report is made possible by general support to CSIS. No direct sponsorship contributed to this report.

Center for Strategic & International Studies
1616 Rhode Island Avenue, NW
Washington, D.C. 20036
202-887-0200 | www.csis.org

Contents

Executive Summary	1
1 Part 1. The Problem	3
<i>Technology Competition Is Talent Competition</i>	3
<i>The Talent Dimensions of the China Challenge</i>	4
<i>China's Domestic Talent Investments Are Paying Off</i>	5
<i>Immigration Is America's Key Asymmetric Talent Advantage</i>	7
<i>Current Policies Risk Squandering America's Immigration Advantage</i>	9
<i>The Emerging Bipartisan Consensus on STEM Immigration as a National Security Priority</i>	9
2 Part 2. The Solutions	12
<i>Converting Consensus into Action</i>	12
<i>Reforming STEM Immigration to Boost National Security</i>	13
<i>Legislative Policy Options</i>	13
<i>Executive Policy Options</i>	14
<i>Ensuring Immigration Reforms Complement Domestic Workforce Investments</i>	15
<i>Legislative Policy Options</i>	15
<i>Executive Policy Options</i>	16
<i>Safeguarding Research and Technology While Maintaining an Open System</i>	16
<i>Legislative Policy Options</i>	17
<i>Executive Policy Options</i>	17
3 Conclusion	19
About the Author	20

Executive Summary

Talent is critical to innovation, and America's deep pool of skilled scientists and engineers is a key component of its technological primacy. But today, for the first time in decades, U.S. leadership is under serious threat. Reaping the fruits of significant long-term investments, China's supply of science, technology, engineering, and mathematics (STEM) talent now rivals that of the United States, both in terms of quantity and quality. Given current trends, it is inevitable that China will overtake the United States in purely domestic terms—if it has not done so already.

The most powerful—and perhaps only—lasting and asymmetric American advantage is its ability to attract and retain international talent, a feat China has not been able to replicate despite extensive efforts. But the U.S. government risks squandering that advantage through poor immigration policy. Without significant reforms to STEM immigration, the United States will struggle to maintain long-term competitiveness and achieve near-term technology priorities such as semiconductor supply chain security, leadership in artificial intelligence (AI), and clean energy innovation.

Part 1 of this paper provides data that supports these claims. Findings include:

- By 2025, China is projected to nearly double annual U.S. STEM PhD output (77,000 versus 40,000 graduates per year, respectively). Counting only domestic U.S. students, China would more than triple American numbers.
- China already far outpaces America in bachelor's and master's graduates; in 2019, China granted 1,886,000 bachelor's degrees and 326,000 master's degrees in STEM fields, compared to 445,000 and 171,000, respectively, for the United States.

- The quality of Chinese education has improved significantly over the last decade, especially at the PhD level. In 2020, 71 Chinese universities ranked in the top 500 globally, up from 23 in 2010. Even when quantity comparisons are limited to graduates from these higher-quality universities, China still graduates more STEM PhDs than the United States does today.
- America remains far more attractive to international scientists and engineers than China. While available data is imperfect, surveys suggest 60 percent of advanced STEM talent based abroad would consider moving to the United States, compared to around 10 percent for China. These figures have changed little over the past decade.
- Immigration policies are harming America’s technology talent pool. Visa issues have contributed to a recent drop in international STEM enrollments. For instance, around 60 percent of U.S.-trained international AI PhDs who left the country after graduating said that immigration issues were relevant to their decision to leave.
- These realities are creating a budding bipartisan consensus that U.S. STEM immigration reform is “a national security imperative,” as the National Security Commission on Artificial Intelligence recently put it. But this consensus has yet to be turned into action.

Part 2 of this report identifies three overarching policy priorities that should be at the center of a twenty-first-century international talent strategy and proposes several concrete legislative and executive policy options for pursuing these priorities (see table below). Along with much-needed domestic investments, these policies would go a long way toward maintaining and expanding America’s technological superiority.

Policy Priorities	Reforming STEM Immigration to Boost National Security	Ensuring Immigration Reforms Complement Domestic Workforce Investments	Safeguarding Research and Technology While Maintaining an Open System
Legislative Policy Options	• Exempt advanced STEM graduates from green card caps. • Create a new green card category for workers in critical and emerging technology fields related to national security. • Create a dedicated and secure student-to-worker pathway.	• Create a dedicated entrepreneur (“start-up”) visa. • Raise visa application fees to fund additional STEM training for domestic students and workers. • Prioritize visa applications in high-demand critical and emerging technology fields.	• Create a new public-private research security partnership. • Prioritize and fund open-source science and technology intelligence.
Executive Policy Options	• Recapture a large number of unused green cards. • Clarify and broaden the scope of the O-1 “extraordinary talent” visa. • Grant “National Interest Waivers” to workers in critical and emerging technology fields.	• Utilize the International Entrepreneur Rule. • Utilize existing immigration authorities related to labor shortages.	• Implement Presidential Proclamation 10043 in a targeted way. • Streamline information sharing across science agencies. • Engage with allies and partners on research security and technology transfer.

Today, the United States still has a tech talent advantage vis-à-vis China; whether it will a decade from now depends, in large part, on U.S. policymakers.

Part 1. The Problem

Technology Competition Is Talent Competition

Technology competition with China is among the central strategic issues facing the U.S. government today. A critical component of this competition involves “talent”—the scientists and engineers working at the frontiers of science and technology.

Talent is core to China’s technological ambitions. Xi Jinping has called talent “the first resource” in China’s drive for “independent innovation,” and Chinese leaders see shortages of high-skilled labor as a key obstacle to their technological ambitions in areas such as semiconductors, AI, and biotechnology.¹ The Chinese Communist Party (CCP) Central Committee wrote in its 2016 National Innovation-Driven Development Strategy that “the essence of being innovation-driven is being talent-driven.”² Huawei founder Ren Zhengfei explains how his strategic priorities are informed by U.S. history: “The strength of the United States as a nation is not land, it’s the talent. What can we learn from the U.S.? Attract talent.”³

Scholars agree with this diagnosis. In a recent review of research on innovation, two prominent economists concluded that “increasing the supply of human capital” is the single best policy tool governments have to sustainably boost technological advancement.⁴ Talent has always played a large role in spurring progress, but it has become especially central since society’s transition toward a “knowledge economy,” where competitiveness depends largely on high-end research and development (R&D) and innovation clusters rather than low-skilled labor and physical production: “Today, the knowledge economy vaults talented individuals to the center of economic performance and the achievement of global prosperity.”⁵

Talent is also critical to the U.S. government’s near-term technological goals. For instance, boosting semiconductor leadership and securing supply chains have been top priorities for both the Trump

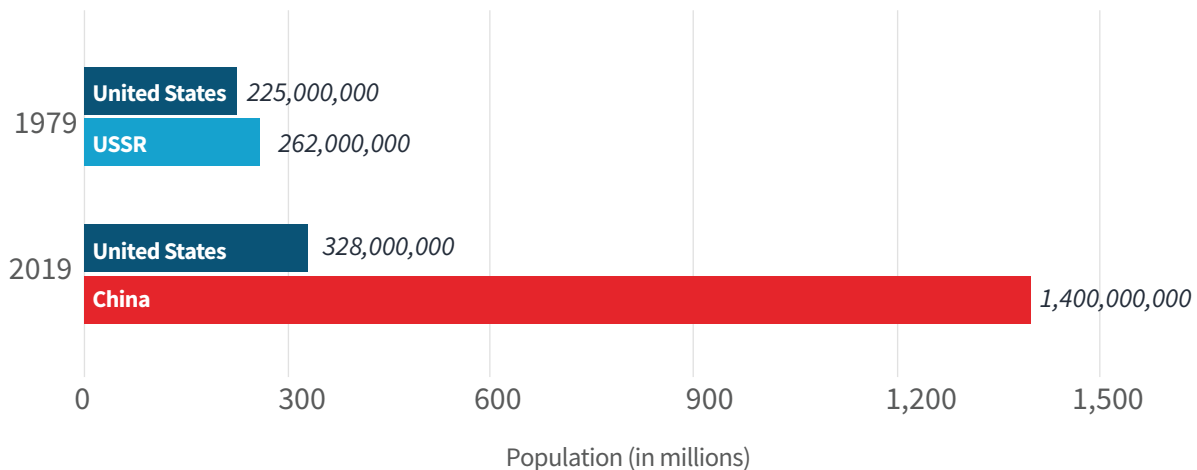
and Biden administrations. Current plans to expand domestic U.S. semiconductor manufacturing capacity, backed by billions in funding, will require tens of thousands of new workers. Semiconductor companies report that labor shortages play a critical role in the current chip shortage and identify workforce investments and immigration reforms as the “number one [policy] change that would help the industry in the near term.”⁶ It is no coincidence that a senior microelectronics professor at Tsinghua, a top Chinese university that recently established a college dedicated to semiconductors, said that “the competition between China and the U.S. is essentially the competition of talent.”⁷

The Talent Dimensions of the China Challenge

In the realm of technology and talent, China is the fiercest challenger the United States has faced since its ascent to great power status. There are two main reasons for this.

The first is China’s sheer scale. America’s last major geopolitical and technological challenger—the Soviet Union—had roughly the same population. China, however, has a population four times the size of the United States (Figure 1). Given this disparity—and the recent successes of China’s educational investments—it is clear that America’s Cold War-era talent strategy, which was centered on domestic workforce investments, is insufficient to meet today’s challenges.

Figure 1: China’s scale means the U.S. government Cold War-era talent strategy is insufficient for today’s context



Source: Census for U.S. data, Demoscope Weekly for USSR data, and National Bureau of Statistics for Chinese data.

The second main reason that today’s talent competition poses an unprecedented challenge for the United States is the increasingly diffuse and competitive nature of the international science and technology ecosystem. The United States dominated global R&D and innovation during much of the Cold War, making it the natural destination for those at the top of their field. Today, it is not nearly as dominant; China alone is projected to exceed U.S. R&D expenditures in 2021.⁸

The United States competes for technical talent not only with China but also with U.S. allies and partners, many of which have recently reformed their immigration systems specifically to attract top technical talent.⁹ Several indicators (reviewed below) suggest that other countries are successfully

poaching large numbers of high-skilled workers who would have otherwise gone to, or stayed in, the United States. America is no longer the default destination for the world's best scientists and engineers.

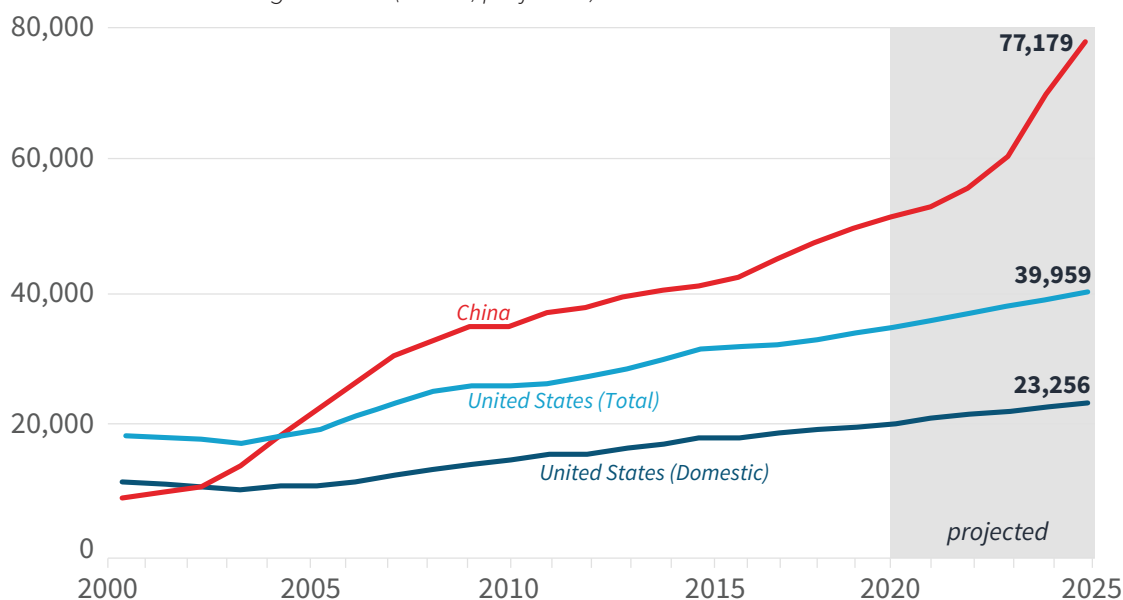
China's Domestic Talent Investments Are Paying Off

Recognizing the importance of talent to technological innovation and independence, China has long prioritized domestic workforce and education investments.¹⁰ A succession of plans—including Project 985, Project 211, and the Double First Class University Plan—have sought to vault Chinese universities into “world-class” status.¹¹ These plans are supported by significant investment; for instance, the budget of the Chinese Ministry of Education doubled between 2012 and 2021.¹²

China's initial development strategy took advantage of the country's large numbers of low-skilled laborers. Today, by contrast, the CCP's policies are focused on converting the country's demographic assets into a high-tech talent advantage. Studies suggest China is succeeding at increasing the quantity and quality of its STEM graduates. In 2019, China graduated 50,000 STEM PhDs, compared to 34,000 in the United States (Figure 2). Based on recent enrollment trends, China is projected to nearly double annual U.S. STEM PhD graduate counts by 2025, with 77,000 compared to 40,000; counting only domestic students, China would more than triple America's projected 23,000 graduates.¹³

Figure 2: China is projected to nearly double U.S. STEM PhD graduates by 2025

Number of STEM Ph.D. graduates (actual/projected)



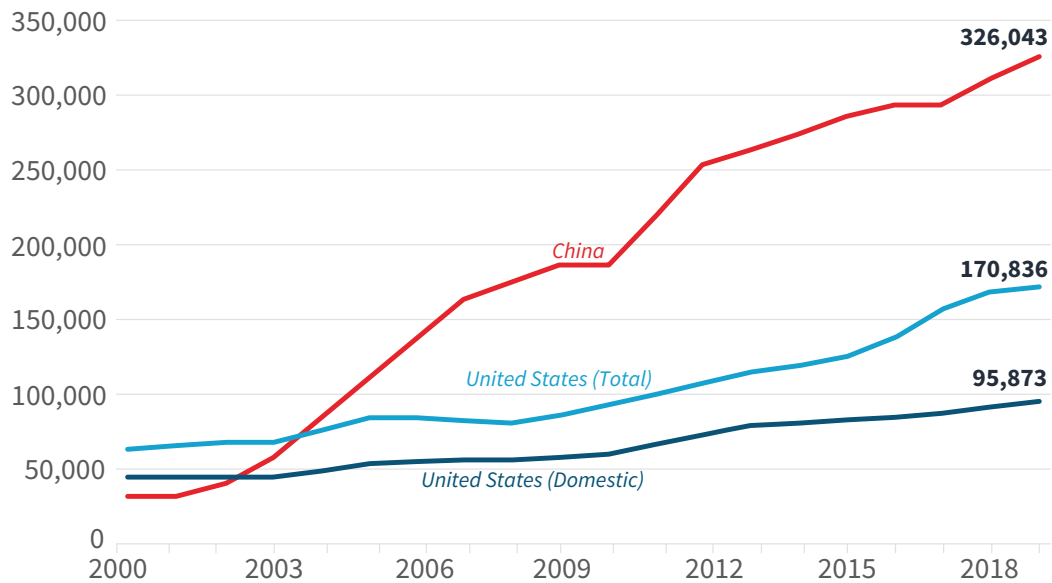
Source: Remco Zwetsloot et al., *China Is Fast Outpacing US STEM PhD Growth* (Washington, DC: Center for Security and Emerging Technology, August 2021), <https://cset.georgetown.edu/publication/china-is-fast-outpacing-u-s-stem-phd-growth/>.

Similar forecasts are not possible for STEM bachelor's and master's graduates due to lack of available data, but past trends show China has already surpassed U.S. universities at these levels as well. In 2019, China graduated 1,886,000 bachelor's students and 326,000 master's students with STEM degrees, compared to 445,000 and 171,000, respectively, in the United States (Figures

3 and 4). Around 45 percent (75,000) of U.S. STEM master's graduates were international students, compared to only 7 percent (31,000) at the bachelor's level; a domestic breakdown is therefore omitted from Figure 4.

Figure 3: China in 2019 already nearly doubled U.S. STEM master's graduates

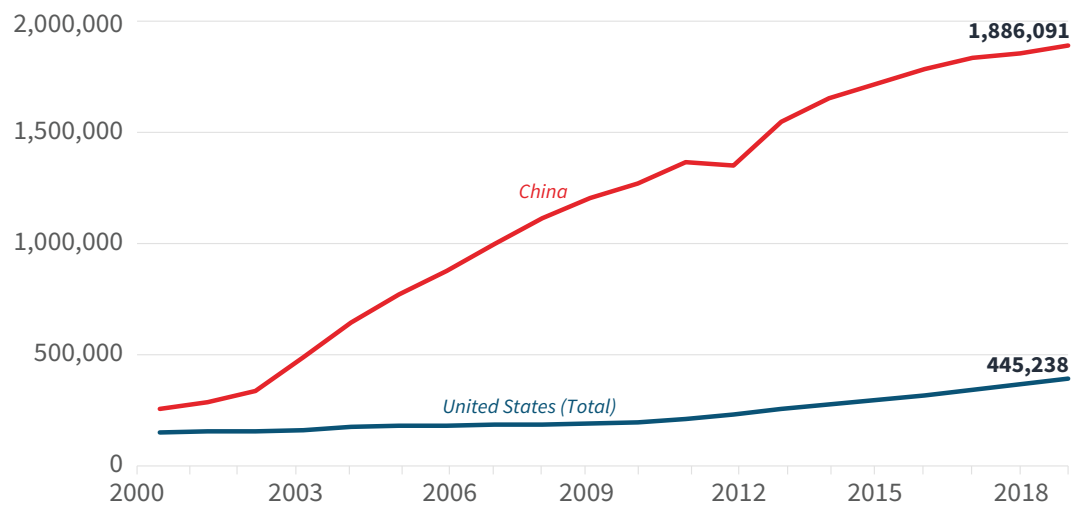
Number of STEM master's graduates



Source: Department of Education IPEDS for U.S. data, Ministry of Education for Chinese data.

Figure 4: China had more than four times as many STEM bachelor's graduates in 2019

Number of STEM bachelor's graduates



Source: Department of Education IPEDS for U.S. data, Ministry of Education for Chinese data.

The quality of Chinese education has also increased, especially at the PhD level. In 2020, 71 Chinese universities ranked in the top 500 globally, up from 23 in 2010. These top-ranked universities graduated an estimated 26,500 STEM PhDs in 2019 (54 percent of all Chinese STEM PhDs), compared to 22,000 STEM PhD graduates from similarly ranked U.S. universities. In other words, even when the comparison is limited to high-quality institutions, China still outproduces the United States in STEM PhD graduates. By contrast, a large majority of China's master's and bachelor's graduates come from lower-quality institutions.¹⁴

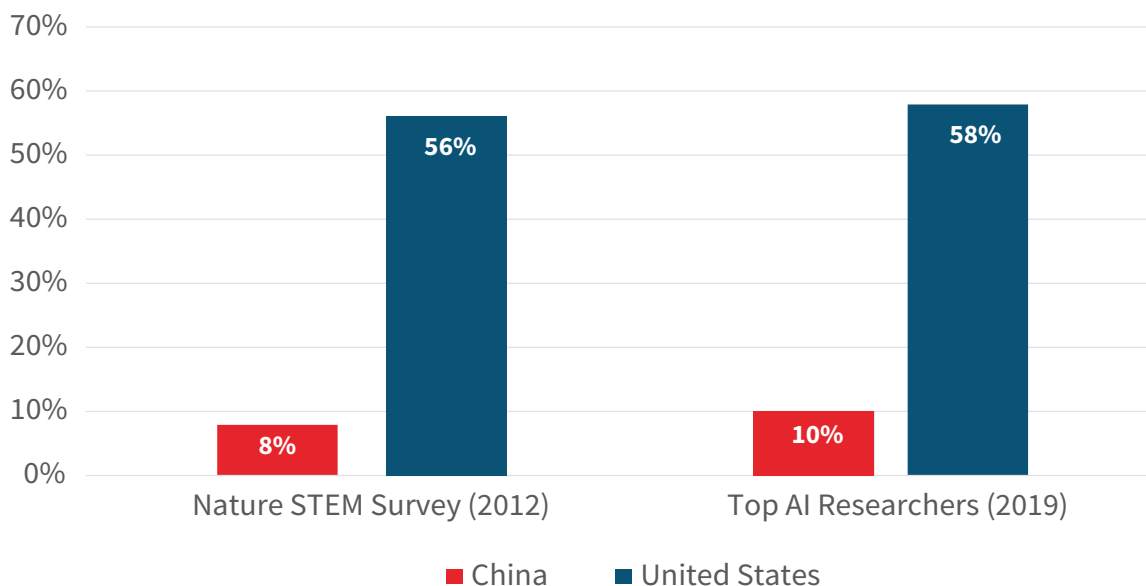
Immigration Is America's Key Asymmetric Talent Advantage

While China has successfully boosted its domestic STEM output, America remains a far more attractive destination for international STEM talent. In this domain of talent competition, China—despite big ambitions and significant investments—has not yet made large-scale gains.

International scientists consistently rate the United States as much more appealing than China. Figure 5 presents the results from two surveys that asked international scientists and engineers where they would consider moving in the near future. Although the surveys were held years apart (2012 versus 2019) and involved different fields (STEM broadly versus AI specifically), the results are consistent: only about 10 percent of international scientists and engineers seemed open to moving to China, compared to nearly 60 percent for the United States.

Figure 5: The United States is much more attractive to international scientists than China

Percent of international scientists who say they would consider moving to . . .



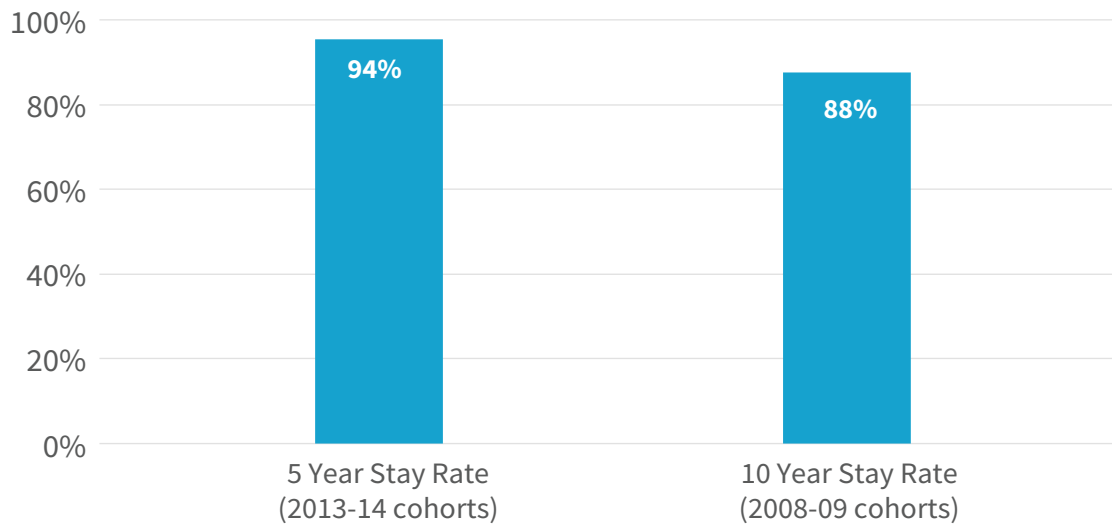
Source: Richard Van Noorden, "Global Mobility: Science on the Move," *Nature*, October 17, 2012, <https://www.nature.com/articles/490326a>; and Remco Zwetsloot et al., "Skilled and Mobile: Survey Evidence of AI Researchers' Immigration Preferences," 2021 AAAI/ACM Conference on AI, Ethics, and Society, <https://arxiv.org/abs/2104.07237>.¹⁵

Data on stay rates among Chinese STEM graduates from U.S. universities suggest that China's struggles also extend to Chinese citizens based abroad. As Figure 6 shows, almost 90 percent of Chinese STEM

PhD students have historically stayed in the United States for at least 10 years after graduating; and at the 10-year mark, nearly all Chinese STEM PhD graduates who remain in the United States are either permanent residents or citizens.¹⁶ Moreover, there is little evidence of recent declines in retention. Surveys that ask STEM PhD graduates whether they intend to stay in the United States for their first post-graduation job find that intention-to-stay rates have consistently been around 85 to 90 percent in recent years (data available up to 2019).¹⁷ The overall scope of China's brain drain problem is significant. One study of international machine learning conference attendees found that the United States may host more than twice as many top Chinese AI researchers than China itself.¹⁸

Figure 6: A large majority of Chinese STEM PhD graduates stay in the United States

Long-term stay rates among Chinese STEM Ph.D. graduates from U.S. universities (as of 2019)



Source: Jack Corrigan, James Dunham, and Remco Zwetsloot, *Long-Term Stay Rates of International STEM PhDs* (Washington, DC: Center for Security and Emerging Technology, forthcoming).

Why has China struggled to attract international talent? Studies suggest that societal and political factors play a major role. For example, in the *Nature* survey, international scientists said they actually considered China to be a more likely source of future breakthrough science than the United States, but political restrictions still made them unwilling to move there.¹⁹ Anecdotal evidence also suggests that many international scientists recruited through Chinese talent programs decide to leave China within a few years—or even months—due to frustrations with political favoritism in resource allocation; repression and censorship of Google Scholar and other platforms; language barriers; and pollution, education, and housing issues.²⁰

To be sure, China still manages to attract some high-profile scientists and engineers. However, because these barriers to recruitment are intimately linked with the CCP's tightening societal controls, it appears unlikely that China will be able to attract and retain international talent at a scale anywhere near the United States. This is why technology and national security leaders have begun calling STEM immigration a “key asymmetric advantage for the United States”—a U.S. strength that China, despite its ability and willingness to invest incredible amounts of money, will not be able to easily replicate.²¹

Immigration as an Asymmetric U.S. Advantage

A decade ago, Joseph Nye asked longtime Singapore leader Lee Kuan Yew whether he thought China would overtake the United States in the twenty-first century. His answer was no, because the United States has long attracted the world's best and brightest, fostering a “diverse culture of creativity,” whereas China will struggle doing so: “China has 1.3 billion people to recruit from domestically, but . . . its Sino-centric culture makes it less creative than the United States, which can draw upon a talent pool of more than 7 billion people.”²²

Current Policies Risk Squandering America's Immigration Advantage

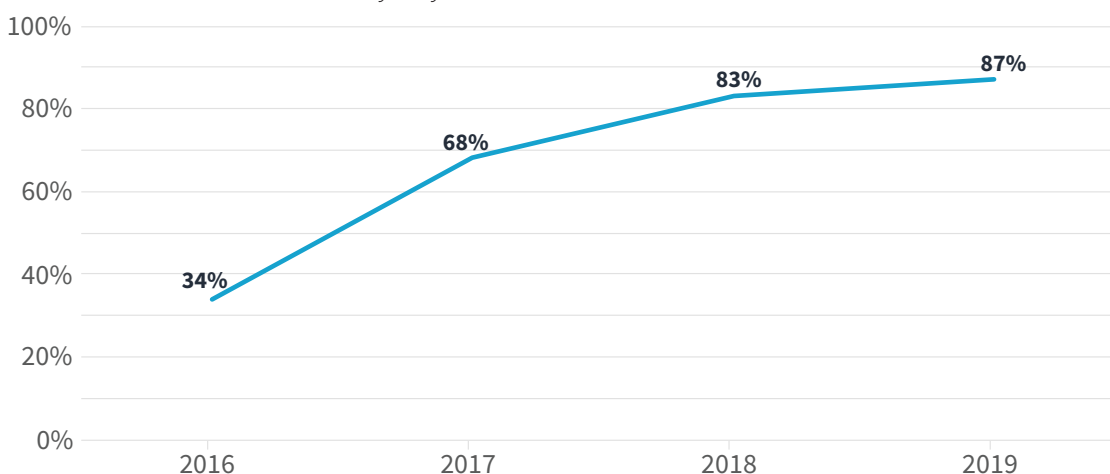
However, the broad appeal of U.S. society is only an asset if the world's best and brightest can actually come and stay in the United States. And on this front, U.S. policy has long been moving in the wrong direction.

Numerical caps on immigration have not materially changed since the 1990s, despite the fact that the U.S. economy and population have grown significantly. These caps limit the annual number of STEM workers that U.S. employers can hire and have caused immense backlogs; in 2020, the employment-based green card backlog—which counts only people whose applications have already been approved by U.S. Citizenship and Immigration Services (USCIS)—exceeded 1.2 million individuals. Under the current system, an Indian STEM PhD holder who receives a job offer today faces a projected wait time of 84 years before their green card application is issued.²³

The negative impact of current U.S. immigration policy is now clearly visible across several indicators. For example, recent declines in international enrollments at U.S. universities appear to be caused at least in part by visa issues (Figure 7). Research also suggests that visa issues and poor immigration prospects disproportionately deter higher-quality students.²⁴

Figure 7: Immigration issues deter international talent from coming to the United States

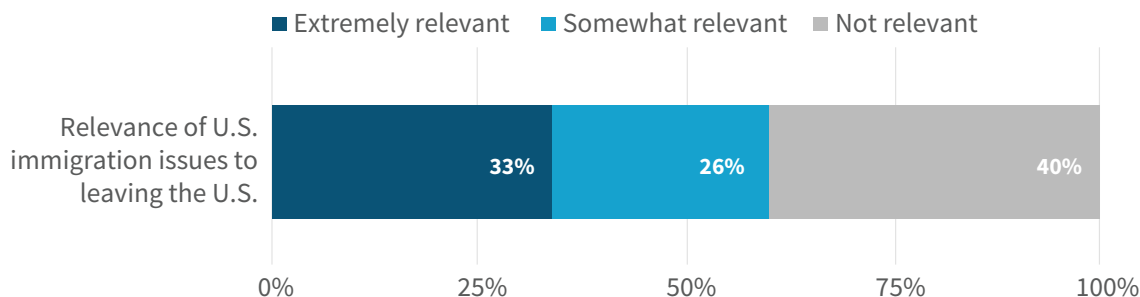
Percent of U.S. universities who say they saw international enrollments decline due to visa issues



Source: “Losing Talent 2020: An Economic and Foreign Policy Risk America Can’t Ignore,” NAFSA: Association of International Educators, March 2020, 6, <https://www.nafsa.org/sites/default/files/media/document/nafsa-losing-talent.pdf>.

Immigration barriers also affect top STEM students who have already come to the United States. In one survey of international AI PhD graduates who left the United States after completing their degrees, more than half report that immigration issues played a role in their decision (Figure 8). Studies like this suggest that STEM immigration reforms could significantly boost U.S. retention of top talent.²⁵

Figure 8: Immigration reforms would help America retain more U.S.-trained STEM talent



Source: Catherine Aiken, James Dunham, and Remco Zwetsloot, “Immigration Pathways and Plans of AI Talent,” Center for Security and Emerging Technology, *Data Brief*, September 2020, <https://cset.georgetown.edu/publication/immigration-pathways-and-plans-of-ai-talent/>.

Other countries are beginning to take advantage of America’s atrophying immigration system, reforming their rules and launching dedicated programs to attract U.S.-based STEM talent.²⁶ Canada, for example, regularly runs billboard campaigns in American technology hubs advertising its opportunities, with slogans like “H-1B Problems? Pivot to Canada.” And these efforts appear to be paying off. Between 2017 and 2019, the number of high-skilled U.S.-based workers who applied for permanent residency in Canada more than doubled; immigration data suggest American companies and universities lost more than 20,000 workers to Canada during these three years alone.²⁷

The Emerging Bipartisan Consensus on STEM Immigration as a National Security Priority

These realities are creating a consensus among U.S. technology and national security leaders that STEM immigration reform is now, as the National Security Commission on Artificial Intelligence recently wrote, a “national security imperative.”²⁸

Chinese Leaders Fear U.S. STEM Immigration Reforms

Chinese leaders understand the extent to which the United States benefits from international talent inflows. They therefore celebrate America's flawed immigration system and fear reforms that would improve U.S. talent attraction and retention.

- Commenting on U.S. retention of Chinese STEM students, the head of the CCP's Central Talent Work Coordination Group has complained that "the number of top talents lost in China ranks first in the world."²⁹
- A state-run consulting firm wrote in an AI policy white paper that U.S. immigration restrictions "have provided China opportunities to bolster its ranks of high-end talent."³⁰
- The deputy editor of *China Daily USA*, a government newspaper, said that expansion of the U.S. employment-based immigration system "would pose a huge challenge for China, which has been making great efforts to attract and retain talent."³¹

Notably, this idea has sparked bipartisan congressional interest. In 2019, the Ronald Reagan Institute launched a task force on the twenty-first-century national security technology workforce, members of which included multiple former and current Republican and Democratic representatives. Its report concluded that "U.S. immigration policies . . . impede the war for talent, often requiring foreign students graduating with high-demand technical degrees to return to their home countries rather than providing pathways for them to stay and contribute to the U.S. [national security innovation base]." To address this problem, it argued "Congress should create a 'National Security Innovation Base Visa' that would encourage appropriately vetted, highly skilled workers to come to the United States or foreign national students graduating with relevant degrees to stay in the United States."³²

The following year, these conclusions were echoed by the Future of Defense Task Force, a group launched by the House Armed Services Committee (HASC) to assess national security priorities for long-term U.S. leadership. Recognizing the importance of talent to great power competition, it looked beyond the traditional boundaries of committee jurisdiction to argue that "the quantifiable success of recent [STEM] immigrants is staggering" and that "immigration policy hinders the U.S.'s ability to attract and retain foreign STEM talent that instead flows to other countries, including competitors," calling on Congress to "aggressively [expand] visas for STEM talent."³³

These arguments were translated into bipartisan legislation by Representatives Jim Langevin and Elise Stefanik, then chair and ranking member on HASC's Subcommittee on Emerging Threats, respectively. Their National Security Innovations Pathway Act creates hundreds of visas (100 in its first fiscal year, up to 500 after five years) for foreign-born talent working in defense-relevant technology areas.³⁴ The act was successfully added as an amendment to the House version of the FY 2021 National Defense Authorization Act, but its Senate counterpart did not include similar language. The provision was ultimately omitted from the bill's reconciled version.

Part 2. The Solutions

Converting Consensus into Action

The emerging bipartisan consensus on STEM immigration must be turned into concrete action before China further expands its domestic advantage and the United States loses out on more international talent. In doing so, policy changes should be made in pursuit of at least three goals: (1) reforming STEM immigration to boost national security, (2) ensuring immigration reforms complement domestic workforce investments, and (3) safeguarding research and technology while maintaining an open system.

Policy Priorities	Reforming STEM Immigration to Boost National Security	Ensuring Immigration Reforms Complement Domestic Workforce Investments	Safeguarding Research and Technology While Maintaining an Open System
Legislative Policy Options	- Exempt advanced STEM graduates from green card caps. - Create a new green card category for workers in critical and emerging technology fields related to national security. - Create a dedicated and secure student-to-worker pathway.	- Create a dedicated entrepreneur (“start-up”) visa. - Raise visa application fees to fund additional STEM training for domestic students and workers. - Prioritize visa applications in high-demand critical and emerging technology fields.	- Create a new public-private research security partnership. - Prioritize and fund open-source science and technology intelligence.
Executive Policy Options	- Recapture a large number of unused green cards. - Clarify and broaden the scope of the O-1 “extraordinary talent” visa. - Grant “National Interest Waivers” to workers in critical and emerging technology fields.	- Utilize the International Entrepreneur Rule. - Utilize existing immigration authorities related to labor shortages.	- Implement Presidential Proclamation 10043 in a targeted way. - Streamline information sharing across science agencies. - Engage with allies and partners on research security and technology transfer.

Part 2 of this paper briefly outlines these goals and lays out several steps that Congress and the executive branch can take to pursue these goals. The lists of policy options are far from exhaustive, but the proposed priorities and policies at least illustrate some of the key components of any twenty-first-century technology talent strategy.

Reforming STEM Immigration to Boost National Security

America's national security depends on the health of the U.S. technology ecosystem, which is often tied to the concept of "STEM"—all of science, technology, engineering, and mathematics. The U.S. government has also recently worked to identify a narrower set of "critical and emerging technology areas" that are especially central to economic and military competitiveness.³⁵ The immigration reforms reviewed below could target STEM fields broadly or focus more specifically on these "critical and emerging" areas (see next section for more discussion).

Immigration policy can also bolster national security by increasing the U.S. government's access to technical talent. From this perspective, it is especially important that policymakers ensure international talent is able to obtain permanent residency and, eventually, citizenship. Focusing on permanent residency also makes sense from a broader competitiveness perspective, as green card caps are the principal bottleneck in the U.S. immigration system. The recommendations below therefore focus primarily on green card reforms.³⁶

LEGISLATIVE POLICY OPTIONS

- **Exempt advanced STEM graduates from green card caps.**³⁷ Currently, the U.S. government can issue 140,000 employment-based green cards per year. Legislation should exempt advanced STEM graduates from those caps.³⁸ The number of exempted individuals will depend on whether it applies to PhDs only or to master's graduates as well; U.S. universities currently graduate around 20,000 international STEM PhD students per year, compared to around 100,000 master's graduates (see Figures 3 and 4 above). If degree holders from non-U.S. universities are included (e.g., anyone in the world with a STEM PhD), the number of individuals eligible for the cap exemption would be significantly higher; exact numbers would depend on the criteria used.
- **Create a new green card category for workers in critical and emerging technology fields related to national security.** If STEM exemptions are infeasible, a more targeted approach could increase the number of available green cards—and speed the path to citizenship—for individuals working in a select set of fields. One model for this approach is the bipartisan National Security Innovations Pathway Act discussed above, which would make available up to 500 new green cards for individuals working in "critical" technology areas.³⁹
- **Create a dedicated and secure student-to-worker pathway.** International graduates from U.S. universities currently account for the majority of new STEM immigrants.⁴⁰ The Optional Practical Training (OPT) program allows F-1 student visa holders to work for a U.S. employer for up to three years after graduation (one year for non-STEM graduates). While it is little-known, OPT is essential for U.S. retention of international STEM graduates and for American firms; more than 1 million STEM graduates have used OPT since 2004, and the chip giant Intel has said that "without OPT, we would be able to hire just 30% of the highly skilled graduates we currently hire."⁴¹ But the program was created through regulation rather than legislation, rendering it insecure; it has been challenged in court and the Trump administration reportedly came close to eliminating the

program. Congress should enshrine OPT (or an equivalent student-to-work pathway) into statute to protect it from such threats.

EXECUTIVE POLICY OPTIONS

- **Recapture a large number of unused green cards.** There are currently hundreds of thousands of “unused” green cards: permanent residency slots that agencies could have issued in prior years but, for a variety of reasons, did not.⁴² Such unused green cards can be “recaptured” and allocated to current applicants. Recapture would significantly reduce green card backlogs and likely help tens of thousands of STEM workers. It is unclear whether recapture can happen without legislation; the American Immigration Lawyers Association argues agencies can “implement [recapture] through administrative means not requiring legislation,” though other legal experts have doubts.⁴³ If further legal assessments show that executive green card recapture is not possible, or only a partial solution, congressional action should be a high priority.⁴⁴
- **Clarify and broaden the scope of the O-1 “extraordinary talent” visa.** The O-1 temporary worker visa is for individuals with “extraordinary ability or achievement” in the sciences (among other fields). The visa is statutorily uncapped and can be renewed indefinitely, which makes it attractive to advanced STEM talent, but immigration lawyers often advise clients against applying due to the uncertainty and administrative burden of the O-1 application and adjudication process. Executive actions could make the O-1 visa more accessible by clarifying which metrics are used to assess eligibility (e.g., what it means to publish in a “major” outlet or do “original work” of “major significance”).⁴⁵ About 10,000 to 15,000 new O-1 visas are currently issued annually; relaxing the criteria, within statutory bounds, could feasibly increase eligibility by thousands of STEM workers per year.
- **Grant “National Interest Waivers” to workers in critical and emerging technology fields.** As part of the “exceptional ability” employment-based green card category (EB-2), for which there are roughly 40,000 slots per year, USCIS is allowed to grant applicants a National Interest Waiver (NIW). The waiver allows companies to sidestep time-consuming recruitment and labor certification (PERM) requirements and workers to sponsor their own application. To qualify for a NIW, applicants must demonstrate that their “proposed endeavor has both substantial merit and national importance” and that they are “well positioned to advance the proposed endeavor.” To encourage the appropriate granting of NIWs to workers in fields that the U.S. government has designated as “critical and emerging technologies,” USCIS could issue public guidance confirming that certain fields, such as AI, are of “national importance.”⁴⁶ This could save thousands of STEM workers per year several months in the application process, reducing administrative burdens and backlogs.

Ensuring Immigration Reforms Complement Domestic Workforce Investments

The U.S. government’s foremost talent policy priority should be domestic workers and students. The National Security Commission on AI called for a “National Defense Education Act II,” with ambitions equal in scope and scale to the U.S. government’s investments made in the wake of Sputnik through the first National Defense Education Act.⁴⁷ Such measures are necessary, but, as the commission recognized, they are not sufficient. Today, the labor needs of the U.S. science and technology system—and the scope of the talent challenge—are simply too great to be met by domestic talent alone.⁴⁸

Fortunately, immigration reforms and domestic workforce investments can complement each other. Fears that international STEM talent mainly “crowds out” domestic students and workers appear exaggerated. A large body of economic evidence suggests that high-skilled immigrants generally have neutral or positive impacts on domestic employment and wage levels and that international students tend to boost domestic enrollments by funding university program expansions.⁴⁹ Recent data on U.S. physics PhD programs suggest that decreases in international applicants did not lead to more domestic enrollment—instead, universities were forced to shrink their programs.⁵⁰ Nonetheless, even small amounts of displacement are cause for concern and could erode support for immigration reforms. The recommendations outlined below would help ensure that STEM immigration measures benefit domestic talent and can be supported by a broad political coalition.

LEGISLATIVE POLICY OPTIONS

- **Create a dedicated entrepreneur (“start-up”) visa.** The U.S. immigration system is designed for employees; most employment-based visa categories require employer sponsorship. This strongly disadvantages entrepreneurs who want to start their own companies, forcing them to abandon their plans, to find awkward workarounds under existing visa categories, or, frequently, to take their companies elsewhere. Those who have found workarounds demonstrate the potential value of immigrant entrepreneurs: more than half of America’s 91 recent “unicorns” (new companies worth \$1 billion or more) had immigrant founders, and, according to one analysis, “immigrant founders have created an average of more than 1,200 jobs per company.”⁵¹ By allowing more foreign talent to start high-technology companies, a “start-up” visa would boost U.S. innovation and create jobs for American workers.⁵² Several legislative templates for such a visa category already exist.⁵³
- **Raise visa application fees to fund additional STEM training for domestic students and workers.** Hundreds of millions in H-1B visa application fees (\$350 million in FY 2019) are already spent on STEM training for domestic students and workers. The funds, which fall outside of the annual appropriations process, are allocated by the National Sciences Foundation and Department of Labor (DOL). Current fee levels were set in 2004 and have not been updated since; simply adjusting the fees for post-2004 inflation would raise them by 30 percent (more than \$100 million). Large companies have also expressed a willingness to pay higher fees. One existing proposal would raise annual income from H-1B fees to \$1 billion, nearly tripling funds for domestic STEM training activities.⁵⁴ The recent National Security Innovations Pathway Act adopted a similar approach by allocating green card application fees to domestic STEM scholarships.
- **Prioritize visas for high-demand critical and emerging technology fields.** Currently, the availability of U.S. visas is tied only indirectly to labor market conditions. Other countries, including Canada and the United Kingdom, more directly integrate economic data into their immigration systems. This is important because not all STEM fields follow the same labor market cycles; talent shortages may exist in some STEM fields or occupations but not in others.⁵⁵ Explicitly prioritizing technology fields and occupations where data suggests that domestic labor supply cannot meet demand would alleviate concerns about immigrants “crowding out” domestic workers. In practice, this could be accomplished by reserving a number of new or existing visa slots for workers in professions deemed by the DOL or other agencies to be facing shortages (see below on related existing authorities). The UK and Canadian systems can serve as inspiration.⁵⁶

EXECUTIVE POLICY OPTIONS

- **Utilize the International Entrepreneur Rule.** Until a legislative “start-up” visa is created, the executive branch could utilize the International Entrepreneur Rule (IER), which grants a period of authorized stay to international entrepreneurs who demonstrate that “their stay in the United States would provide a significant public benefit through their business venture.”⁵⁷ If the IER were fully utilized, around 3,000 additional entrepreneurs could come to the United States per year according to Department of Homeland Security estimates. A recent study projects that, in this scenario, the IER program would create between 100,000 and 300,000 jobs over 10 years. The study also discusses several specific executive actions that would promote IER utilization.⁵⁸ The Biden administration already appears to be taking steps in this direction.
- **Utilize existing immigration authorities related to labor shortages.** Under current law, the DOL can designate occupations as suffering from a labor shortage, meaning DOL has “determined there are not sufficient U.S. professionals who are able, willing, qualified and available” for work and that hiring foreign professionals therefore “will not adversely affect the wages and working conditions of U.S. professionals similarly employed.”⁵⁹ If an occupation is added to the shortage list (called “Schedule A”), companies that want to sponsor workers for employment-based green cards (specifically EB-2 and EB-3) are exempt from time-consuming recruitment and labor certification (PERM) requirements. Schedule A authorities were used extensively during the Cold War to recruit talent in high-priority areas such as aeronautical and electrical engineering.⁶⁰ However, there are currently only two occupations on the list (physical therapists and nurses), and the list has not been kept up to date, despite indications of shortages in critical fields such as AI.⁶¹ Prioritizing STEM immigrants in occupations with shortages can both boost national security and help alleviate concerns about immigrants “crowding out” domestic workers.

Safeguarding Research and Technology while Maintaining an Open System

America’s open science and technology ecosystem confers many advantages, including the foreign talent that is attracted to its shores, the networks that keep U.S. scientists and engineers at the cutting edge, and the science and technology intelligence that can be gained through international collaborations. In the context of talent competition with China, one particularly salient advantage is the “brain drain dilemma” that U.S. openness creates for the CCP: Chinese students remain in the United States at much higher rates than Chinese leaders would like.⁶²

But openness also poses inevitable risks. For example, government officials are concerned some Chinese students and scientists wittingly or unwittingly work as “nontraditional collectors” in pursuit of the CCP’s technology priorities.⁶³ Such risks can and must be dealt with while simultaneously maintaining the fundamental openness of the system. Striking this balance requires targeted reforms on both the domestic and international levels.

LEGISLATIVE POLICY OPTIONS

- **Create a new public-private research security partnership.** Recent research security measures have been focused on federally funded research, for example, investigating the accuracy of information provided in grant applications. However, only around 20 percent of U.S. R&D is federally funded,

and current efforts suffer from limited information, authorities, and trust.⁶⁴ To address these issues, experts have proposed a public-private partnership—an “independent research security clearinghouse” that would be “empowered . . . but not run by the government”—based on successful models from the cybersecurity world, such as the National Cyber Forensics and Training Alliance.⁶⁵ This institution could collect and synthesize non-public and public data, disseminate threat findings and best practices, and serve as a non-punitive forum where stakeholders—whether federally funded or not—could share concerns and receive tailored advice. Congress should provide the resources and authorities necessary for relevant federal agencies to enter into research security partnerships with academia, industry, and other R&D funders and performers.

- **Prioritize and fund open-source science and technology intelligence.** Much of the world’s useful information on scientists, engineers, and their activities is available in the public domain (“open source”). Several recent Department of Justice indictments related to research security were based in large part on open-source intelligence (OSINT), for example, photos of Chinese scientists in military uniform publicly available on the Chinese internet.⁶⁶ Yet U.S. intelligence agencies have been criticized for prioritizing “marquee, classified collection activity” and “consistently marginaliz[ing] OSINT for decades.”⁶⁷ By one expert estimate, China employs roughly 100,000 OSINT-focused science and technology analysts in its intelligence enterprise, compared to perhaps 100 in the U.S. government.⁶⁸ To address these concerns and imbalances, Congress should support a new, largely open-source-based National Science and Technology Analysis Center (NSTAC), of which one focus area would be international talent flows and risk assessments.⁶⁹

EXECUTIVE POLICY OPTIONS

- **Implement Presidential Proclamation 10043 in a targeted way.** In May 2020, the Trump administration issued a proclamation barring Chinese graduate students and researchers from receiving visas if they were ever affiliated with Chinese institutions that “implement or support” China’s “military-civil fusion strategy.”⁷⁰ The proclamation tackles an important problem but also left many key terms undefined, causing significant uncertainty about its scope and eventual impact. Some worried it could “be interpreted as an effective ban on Chinese students.”⁷¹ The proclamation should be implemented in a targeted manner, with clear guidance for government officials on how to assess individual-level risk factors. Implementation will also benefit from greater science and technology intelligence resources (discussed above).
- **Streamline information sharing across science agencies.** To combat conflicts of commitment and deception, one of the main focus areas of recent research security efforts has been on scrutinizing undisclosed ties to foreign governments; the National Institutes of Health alone investigated hundreds of researchers since 2016, more than 50 of whom reportedly lost their jobs.⁷² Greater integration of U.S. science agencies’ data systems would allow information and potential red flags (e.g., on grantees, contractors, and funding applications) to be shared across the major science funders. This would help enforce research integrity rules while also benefiting researchers and universities by reducing administrative requirements.⁷³ The White House’s Joint Committee on the Research Environment (JCORE) should continue prioritizing these reforms.⁷⁴
- **Engage with allies and partners on research security and technology transfer.** Most of the world’s cutting-edge R&D takes place outside of the United States. If the United States imposes restrictions unilaterally, Chinese technology acquisition would simply shift elsewhere, decreasing U.S. competitiveness without meaningfully slowing China’s technological growth. These problems

are familiar from areas such as export and investment controls, where the United States was a key player in building multilateral structures for policy coordination and intelligence sharing. Several proposals now exist for similar initiatives in the domain of research security and technology transfer.⁷⁵ These conversations should include the top non-U.S. destinations for Chinese students today, all of which—Australia, the United Kingdom, Canada, Japan, and South Korea, as well as several European countries—are U.S. allies.⁷⁶

Conclusion

The U.S. government should urgently take steps to reform its STEM immigration policies, ensure these immigration reforms complement ambitious domestic investments, and safeguard research and technology while maintaining the openness of America's science and technology ecosystem. China is reaping the rewards of longstanding domestic STEM investments, with rapid gains in both quantity and quality that show no signs of slowing down. Meanwhile, problems with the U.S. immigration system are worsening and increasingly deterring international talent. America can no longer assume it is the default destination for the world's best scientists and engineers.

Talent is arguably the single most important ingredient driving innovation. If the United States lost its international talent advantage, it would be a big nail in the coffin of American technological leadership. Fortunately, U.S. policymakers have several policy levers available to them to reverse current trends—but this is good news only to the extent that they use those levers. The time for action is now.

About the Author

Remco Zwetsloot is a trustee fellow in the International Security Program at the Center for Strategic and International Studies (CSIS) and a research fellow at the Center for Security and Emerging Technology (CSET). His areas of expertise include the U.S. technology workforce, STEM immigration, research security, and technology competition with China. His writing has been published in a variety of popular, policy, and academic outlets, including the *Wall Street Journal*, *Washington Post*, and *Foreign Affairs*, and his work has been cited in White House reports, congressional testimony, the National Security Commission on Artificial Intelligence, and publications including the *New York Times*, *Science*, and *The Atlantic*. He is also a research affiliate of the University of Oxford's Center for the Governance of AI, where he is completing his PhD in international relations. He holds a master's degree in political science from Yale University, a master's degree in international relations from Oxford, and a bachelor's degree in social science from University College Roosevelt.

Endnotes

- 1 Remco Zwetsloot, *China's Approach to Tech Talent Competition* (Washington, DC: Brookings Institution, April 2020), 1–2, <https://www.brookings.edu/research/chinas-approach-to-tech-talent-competition/>.
- 2 “Outline of the National Innovation-Driven Development Strategy,” Central Committee of the Communist Party of China and the State Council of the People's Republic of China, May 19, 2016, trans. Center for Security and Emerging Technology, https://cset.georgetown.edu/wp-content/uploads/t0076_innovation_driven_development_strategy_EN.pdf.
- 3 Cheng Yu, “Huawei Fat Paychecks Premiumize Tech Talent,” *China Daily*, July 15, 2021, <https://global.chinadaily.com.cn/a/202107/15/WS60ef8f30a310efa1bd662052.html>.
- 4 Nicholas Bloom, John Van Reenen, and Heidi Williams, “A Toolkit of Policies to Promote Innovation,” *Journal of Economic Perspectives* 33, no. 3 (2019), doi:10.1257/jep.33.3.163.
- 5 William Kerr, *The Gift of Global Talent: How Migration Shapes Business, Economy, and Society* (Stanford, CA: Stanford University Press), 4.
- 6 “The Current Sentiment of the Global Electronics Manufacturing Supply Chain,” IPC, September 2021, <https://emails.ipc.org/links/0921Current-Sentiment-GEMSC.pdf>; and “SIA Workforce Roundtable Summary Report,” Semiconductor Industry Association, March 16, 2018, https://www.semiconductors.org/wpcontent/uploads/2018/06/Roundtable_Summary_Report_-_FINAL.pdf. For an analysis of future workforce needs, see Will Hunt, “Workforce Implications of CHIPS for America Manufacturing Incentives,” (unpublished paper, April 2021). For background, see Will Hunt and Remco Zwetsloot, *The Chipmakers: US Strengths and Priorities for the High-End Semiconductor Workforce* (Washington, DC: Center for Security and Emerging Technology, September 2020), <https://cset.georgetown.edu/publication/the-chipmakers-u-s-strengths-and-priorities-for-the-high-end-semiconductor-workforce/>.
- 7 Wei Sheng, “Where Firms Are Looking to Fill China's Chip Talent Gap,” Technode, April 1, 2021, <https://technode.com/2021/04/01/where-firms-are-looking-to-fill-chinas-chip-talent-gap/>.
- 8 American Academy of Arts and Sciences, *The Perils of Complacency: America at a Tipping Point in Science and Engineering* (Cambridge, MA: 2020), 9–10, [https://www.amacad.org/sites/default/files/publication/downloads/Perils-of-Complacency-](https://www.amacad.org/sites/default/files/publication/downloads/Perils-of-Complacency.pdf)

- 9 Tina Huang and Zachary Arnold, *Immigration Policy and the Global Competition for AI Talent* (Washington, DC: Center for Security and Emerging Technology, June 2020), <https://cset.georgetown.edu/publication/immigration-policy-and-the-global-competition-for-ai-talent/>.
- 10 For a detailed history of China's education and workforce investments, see Denis Simon and Cong Cao, *China's Emerging Technological Edge: Assessing the Role of High-End Talent* (Cambridge, UK: Cambridge University Press, 2009).
- 11 Guanzi Shen, "Building World-Class Universities in China: From the View of National Strategies," Global University Network for Innovation, May 5, 2018, <http://www.guninetwork.org/articles/building-world-class-universities-china-view-national-strategies>.
- 12 Ryan Fedasiuk, Alan Omar Loera Martinez, and Anna Puglisi, *China's Support for Universities, Science, and Technology* (Washington, DC: Center for Security and Emerging Technology, forthcoming).
- 13 For details on data and methodology, see Remco Zwetsloot et al., *China Is Fast Outpacing US STEM PhD Growth* (Washington, DC: Center for Security and Emerging Technology, August 2021), <https://cset.georgetown.edu/publication/china-is-fast-outpacing-u-s-stem-phd-growth/>.
- 14 While 54 percent of China's PhDs graduate from top-ranked institutions, lower shares do so at the master's and bachelor's levels (30 percent and 7 percent, respectively, in 2019). For sources and methodological details, see Jack Corrigan and Simon Rodriguez, *Ranking U.S. and Chinese Universities* (Washington, DC: Center for Security and Emerging Technology, forthcoming).
- 15 Note that the question was phrased somewhat differently across the two surveys: one (*Nature*) simply asked respondents "Would you consider moving here?," whereas the other asked more specifically which countries respondents had a greater than 25 percent chance of moving to within the next three years.
- 16 Jack Corrigan, James Dunham, and Remco Zwetsloot, *Long-Term Stay Rates of International STEM PhDs* (Washington, DC: Center for Security and Emerging Technology, forthcoming).
- 17 Remco Zwetsloot, Jacob Feldgoise, and James Dunham, *Trends in U.S. Intention-to-Stay Rates of International PhD Graduates Across Nationality and STEM Fields* (Washington, DC: Center for Security and Emerging Technology, April 2020), <https://cset.georgetown.edu/publication/trends-in-u-s-intention-to-stay-rates-of-international-ph-d-graduates-across-nationality-and-stem-fields/>. Data included in that paper goes up to 2018; the same data is now available for 2019 and shows no notable trend changes (checked by author, unpublished).
- 18 Joy Dantong Ma, "China's AI Talent Base Is Growing, and then Leaving," MacroPolo, July 30, 2019, <https://macropolo.org/chinas-ai-talent-base-is-growing-and-then-leaving/>.
- 19 In the 2012 survey, 59 percent of respondents said they expected China to have the greatest scientific impact by 2020, compared to 36 percent for the United States. Making China less attractive, 93 percent of respondents identified "authoritarian political system and restricted freedom" as a barrier to migration (the highest proportion of any factor); see Van Noorden, "Global Mobility."
- 20 For several examples, see Remco Zwetsloot and Dahlia Peterson, "The US-China Tech Wars: China's Immigration Disadvantage," *The Diplomat*, December 31, 2019, <https://thediplomat.com/2019/12/the-us-china-tech-wars-chinas-immigration-disadvantage/>.
- 21 "Asymmetric Competition: A Strategy for China and Technology," China Strategy Group, Fall 2020, <https://www.documentcloud.org/documents/20463382-final-memo-china-strategy-group-axios-1>.
- 22 Joseph Nye, "A Key to America's Power," CNN, June 12, 2015, <https://www.cnn.com/2015/06/12/opinions/nye-immigration-boosts-american-future/index.html>.
- 23 David J. Bier, "Employment-Based Green Card Backlogs Hits 1.2 Million in 2020," CATO Institute, November 20, 2020, Table 1, <https://www.cato.org/blog/employment-based-green-card-backlog-hits-12-million-2020>.
- 24 Takeo Kato and Chad Sparber, "Quotas and Quality: The Effect of H-1B Visa Restrictions on the Pool of Prospective Under-

- graduate Students from Abroad,” *Review of Economics and Statistics* 95, no. 1 (March 2013), doi:10.1162/REST_a_00245.
- 25 See also Shulamit Kahn and Megan MacGarvie, “The Impact of Permanent Residency Delays for STEM PhDs: Who Leaves and Why,” National Bureau of Economic Research, October 2018, <http://www.nber.org/papers/w25175>; and other studies reviewed in Remco Zwetsloot et al., *Keeping Top AI Talent in the United States* (Washington, DC: Center for Security and Emerging Technology, December 2019), 10–12, <https://cset.georgetown.edu/publication/keeping-top-ai-talent-in-the-united-states/>.
 - 26 Huang and Arnold, *Immigration Policy and the Global Competition for AI Talent*.
 - 27 Zachary Arnold, “Canada’s Immigration System Increasingly Draws Talent from the United States,” Center for Security and Emerging Technology, *Data Brief*, July 2020, <https://cset.georgetown.edu/publication/canadas-immigration-system-increasingly-draws-talent-from-the-united-states/>.
 - 28 National Security Commission on Artificial Intelligence, *Final Report* (Washington, DC: 2021), 175, <https://www.nsc.gov/2021-final-report/>.
 - 29 Remco Zwetsloot and Zachary Arnold, “Chinese Students Are Not a Fifth Column,” *Foreign Affairs*, April 23, 2021, <https://www.foreignaffairs.com/articles/united-states/2021-04-23/chinese-students-are-not-fifth-column>.
 - 30 Ibid.
 - 31 Ibid.
 - 32 Reagan Institute, *The Contest for Innovation: Strengthening America’s National Security Innovation Base in an Era of Strategic Competition* (Chicago: December 2019), 23–24, https://www.reaganfoundation.org/media/355297/the_contest_for_innovation_report.pdf.
 - 33 House Armed Services Committee, *Future of Defense Task Force Report 2020* (Washington, DC, September 2020), 17, 56, 57, <https://armedservices.house.gov/2020/9/future-of-defense-task-force-releases-final-report>.
 - 34 U.S. Congress, House, *National Security Innovation Pathway Act*, H.R.7256, 116th Cong., 2nd sess., Introduced June 18, 2020, <https://www.congress.gov/bill/116th-congress/house-bill/7256>.
 - 35 The first list of “critical and emerging technologies,” which included 20 technology areas, was released in October 2020. The list is supposed to be reviewed and updated through an interagency process on an annual basis; see White House, *National Strategy for Critical and Emerging Technologies* (Washington, DC: October 2020), A1, <https://trumpwhitehouse.archives.gov/wp-content/uploads/2020/10/National-Strategy-for-CET.pdf>.
 - 36 This section will briefly outline a small number of high-priority reforms; for further background on the U.S. STEM immigration system more broadly, see e.g., Arnold et al., *Immigration Policy and the US AI Sector* (Washington, DC: Center for Security and Emerging Technology, September 2019), <https://cset.georgetown.edu/publication/immigration-policy-and-the-u-s-ai-sector/>; and Zwetsloot et al., *Keeping Top AI Talent in the United States*, 19–24. For discussion on other potential STEM and high-skill immigration reforms, see e.g., Sari Pekkala Kerr and William Kerr, “Immigration Policy Levers for US Innovation and Startups,” National Bureau of Economic Research, November 2020, <https://www.nber.org/system/files/chapters/c14424/c14424.pdf>.
 - 37 Two other cap-related measures would also significantly benefit advanced STEM degree holders, even though they are not STEM-specific. First, around half of available employment-based green cards currently go to spouses and children (“derivative beneficiaries”), who are counted toward the cap. Exempting derivative beneficiaries from the cap would thus effectively double the number of available employment-based green cards. Second, legislation could simply raise the caps altogether, either to a new fixed number or pegged to a moving indicator that would allow the cap to vary based on economic conditions (e.g., the U.S. population or GDP).
 - 38 This idea differs from proposals to “staple” a green card to international students’ STEM degrees. A cap exemption would not automatically grant graduates permanent residency, as it still requires students to obtain employer sponsorship. This “exemption” model, by tying immigration prospects to employer demand and employability, reduces the risk of domestic displacement (see below) and “diploma mills” (low-quality universities created for the purpose of buying educational and immigration credentials).

- 39 U.S. Congress, House, *National Security Innovation Pathway Act*. Another potential model is the Military Accessions Vital to National Interest (MAVNI) program, created in 2008, which allows the DOD to hire nonimmigrant staff with skills critical to U.S. national interests (e.g., languages) who are then placed on a fast-track to citizenship.
- 40 Zwetsloot et al., *Keeping Top AI Talent in the United States*, 22–23.
- 41 Zachary Arnold and Remco Zwetsloot, “Optional Practical Training,” Center for Security and Emerging Technology, September 2020, <https://cset.georgetown.edu/publication/optional-practical-training/>.
- 42 Jeremy L. Neufeld, Lindsay Milliken, and Doug Rand, *Stop the Incinerator: The High Cost of Green Card Slots Going Unused and the Benefits of Recapturing Them* (Washington, DC: Niskanen Center, June 2021), <https://www.niskanencenter.org/stop-the-incinerator-the-high-cost-of-green-card-slots-going-unused-and-the-benefits-of-recapturing-them/>.
- 43 “AILA Recommendations for the Future of Immigration,” American Immigration Lawyers Association, January 2021, 15, <https://www.aila.org/infonet/a-vision-for-america-as-a-welcoming-nation>.
- 44 For discussions of why it is desirable for recapture to happen legislatively and how congressional action could be more significant in scope than executive recapture, see Neufeld, Milliken, and Rand, *Stop the Incinerator*, and “Green Card Recapture and Reform Would Reduce Immigration Backlogs,” FWD.us, April 13, 2021, <https://www.fwd.us/news/green-card-recapture/>.
- 45 Doug Rand and Lindsay Milliken, “How the Executive Branch Can Streamline US Immigration Options for AI Talent,” *NYU Journal of Legislation and Public Policy*, April 9, 2021, <https://nyujlpp.org/quorum/rand-milliken-winning-global-race-artificial-intelligence/>.
- 46 Ibid.
- 47 National Security Commission on AI, *Final Report*, 175, 421–24.
- 48 On the fast-growing labor demands in high-technology fields, see e.g., Nicholas Bloom et al., “Are Ideas Getting Harder to Find?,” *American Economic Review* 110, no. 4 (April 2020), doi:10.1257/aer.20180338. This study found, for example, that achieving an equivalent unit of technological progress in semiconductors takes 18 times as many researchers today as it did in the early 1970s.
- 49 For discussions of the evidence on enrollment impacts, see Kevin Shih, “Do International Students Crowd-Out or Cross-Subsidize Americans in Higher Education?,” *Journal of Public Economics* 156 (2017), 171, doi:10.1016/j.jpubeco.2017.10.003; and John Bound et al., “The Globalization of Postsecondary Education: The Role of International Students in the US Higher Education System,” *Journal of Economic Perspectives* 35, no. 1 (Winter 2021), doi:10.1257/jep.35.1.163. On labor market impacts, see Alex Nowrasteh, “Don’t Ban H-1B Workers: They Are Worth Their Weight in Innovation,” CATO Institute, May 14, 2020, <https://www.cato.org/blog/dont-ban-h-1b-workers-they-are-worth-their-weight-patents>; and National Academies, *The Economic and Fiscal Consequences of Immigration* (Washington, DC: 2017), 249, <https://www.nationalacademies.org/our-work/economic-and-fiscal-impact-of-immigration>.
- 50 American Physical Society, *Building America’s STEM Workforce: Eliminating Barriers and Unlocking Advantages* (College Park, MD: January 2021), 11, <https://www.aps.org/policy/analysis/upload/Building-America-STEM-workforce.pdf>.
- 51 Stuart Anderson, “Immigrants and Billion-Dollar Companies,” National Foundation for American Policy, October 2018, <https://nfap.com/wp-content/uploads/2018/10/2018-BILLION-DOLLAR-STARTUPS.NFAP-Policy-Brief.2018.pdf>.
- 52 One analysis of past visa start-up proposals estimated that they could create between 1 million and 3.2 million jobs in the United States over a decade; see Stuart Anderson, “Startup Visa Proposals and Job Creation,” National Foundation for American Policy, March 2016, <http://nfap.com/wp-content/uploads/2016/03/NFAP-Policy-Brief.Analysis-of-Startup-Visa-Proposals1.pdf>.
- 53 See, e.g., U.S. Congress, Senate, *Startup Act*, S.328, 116th Cong., 2nd sess., Introduced February 4, 2019, <https://www.congress.gov/bill/116th-congress/senate-bill/328>; and U.S. Congress, House, *LIKE Act*, H.R. 4681, 116th Cong., 2nd sess., July 26, 2021, <https://www.congress.gov/bill/117th-congress/house-bill/4681>.

- 54 Lindsay Milliken and Doug Rand, How to Expand H-1B Fees for Innovative Workforce Training and Inclusive Graduate STEM Education,” Day One Project, June 2021, <https://www.dayoneproject.org/post/building-an-evergreen-1-billion-fund-for-science-and-technology-career-advancement>.
- 55 For an assessment of AI shortages that finds variation across occupations, see Diana Gehlhaus and Ilya Rakhovsky, *US AI Workforce: Labor Market Dynamics* (Washington, DC: Center for Security and Emerging Technology, April 2021), <https://cset.georgetown.edu/publication/u-s-ai-workforce/>. For more background (from different perspectives) on STEM shortage debates in the United States more broadly, see Michael Teitelbaum, *Falling Behind?: Boom, Bust, and the Global Race for Scientific Talent* (Princeton, NJ: Princeton University Press, 2014); and Adams Nager and Robert D. Atkinson, *Debunking the Top Ten Myths Against High-Skilled Immigration* (Washington, DC: Information Technology & Innovation Foundation, April 2015), <https://itif.org/publications/2015/04/20/debunking-top-ten-arguments-against-high-skilled-immigration>.
- 56 Rand and Milliken, “How the Executive Branch Can Streamline US Immigration Options for AI Talent.”
- 57 “International Entrepreneur Parole,” U.S. Customs and Immigration Services, <https://www.uscis.gov/humanitarian/humanitarian-parole/international-entrepreneur-parole>.
- 58 Caleb Watney, Lindsay Milliken, and Doug Rand, “Long Live the International Entrepreneur Rule: An Opportunity to Boost Jobs and Economic Growth Is Hiding in Plain Sight,” Progressive Policy Institute, February 2021, <https://innovationfrontier.org/long-live-the-international-entrepreneur-rule/>.
- 59 Rand and Milliken, “How the Executive Branch Can Streamline US Immigration Options for AI Talent.”
- 60 Lindsay Milliken, “A Brief History of Schedule A: The United States’ Forgotten Shortage Occupation List,” *University of Chicago Law Review Online*, September 22, 2020, <https://lawreviewblog.uchicago.edu/2020/09/22/milliken-schedule-a/>.
- 61 Rand and Milliken, “How the Executive Branch Can Streamline US Immigration Options for AI Talent.”
- 62 National Security Commission on AI, *Final Report*, 177; and Remco Zwetsloot, *US-China STEM Talent ‘Decoupling’: Background, Policy, Impact* (Laurel, MD: Johns Hopkins University Applied Physics Lab, October 2020), 18–20, <https://www.jhuapl.edu/assessing-us-china-technology-connections/publications>.
- 63 FBI Director Christopher Wray quoted in, for example, Elizabeth Redden, “The Chinese Student Threat?,” *Inside Higher Ed*, February 15, 2018, <https://www.insidehighered.com/news/2018/02/15/fbi-director-testifies-chinese-students-and-intelligence-threats>.
- 64 Melissa Flagg and Zachary Arnold, *A New Institutional Approach to Research Security in the United States: Defending a Diverse R&D Ecosystem* (Washington, DC: Center for Security and Emerging Technology, January 2021), <https://cset.georgetown.edu/publication/a-new-institutional-approach-to-research-security-in-the-united-states/>; for more background, see Lisa Ide et al., *Improper Influence in Federally Funded Fundamental Research* (McClean, VA: MITRE Corporation, December 2020), <https://www.mitre.org/publications/technical-papers/improper-influence-in-federally-funded-fundamental-research>.
- 65 Flagg and Arnold, *A New Institutional Approach to Research Security*, 14–17. The NCFTA is “a nonprofit partnership between private industry, government, and academia for the sole purpose of providing a neutral, trusted environment that enables two-way collaboration and cooperation to identify, mitigate, and disrupt cyber crime.’ Working from unclassified office space near Carnegie Mellon University, NCFTA’s team, which includes private-sector analysts, federal investigators, and cybersecurity scholars, has prevented billions of dollars in losses and referred thousands of cases to law enforcement” (15).
- 66 Nidhi Subbaraman, “US Investigations of Chinese Scientists Expand Focus to Military Ties,” *Nature*, September 3, 2020, <https://www.nature.com/articles/d41586-020-02515-x>.
- 67 Tarun Chhabra et al., “Open-Source Intelligence for S&T Analysis,” Center for Security and Emerging Technology, September 2020, <https://cset.georgetown.edu/publication/open-source-intelligence-for-st-analysis/>.
- 68 William Hannas and Huey-Meei Chang, *China’s STI Operations: Monitoring Foreign Science and Technology Through Open Sources* (Washington, DC: Center for Security and Emerging Technology, January 2021), 43, <https://cset.georgetown.edu/publication/chinas-sti-operations/>.
- 69 Chhabra et al., “Open-Source Intelligence for S&T Analysis”; for a detailed sketch, see Hannas and Chang, *China’s STI Operations*, 41–43.

- 70 “Suspension of Entry as Nonimmigrants of Certain Students and Researchers From the People’s Republic of China,” *Federal Register*, 85 FR 34353, June 4, 2020, <https://www.federalregister.gov/documents/2020/06/04/2020-12217/suspension-of-entry-as-nonimmigrants-of-certain-students-and-researchers-from-the-peoples-republic>.
- 71 Remco Zwetsloot, Emily Weinstein, and Ryan Fedasiuk, *Assessing the Scope of US Visa Restrictions on Chinese Students* (Washington, DC: Center for Security and Emerging Technology, February 2021), <https://cset.georgetown.edu/publication/assessing-the-scope-of-u-s-visa-restrictions-on-chinese-students/>.
- 72 Jeffrey Mervis, “‘Has it Peaked? I Don’t Know.’ NIH Official Details Foreign Influence Probe,” *Science*, June 22, 2020, <https://www.science.org/news/2020/06/has-it-peaked-i-don-t-know-nih-official-details-foreign-influence-probe>. For further background on these efforts, see JASON, *Fundamental Research Security* (McClean, VA: MITRE Corporation, December 2019), https://www.nsf.gov/news/special_reports/jasonsecurity/JSR-19-2IFundamentalResearchSecurity_12062019FINAL.pdf; and U.S. Senate Permanent Subcommittee on Investigations, *Threats to the U.S. Research Enterprise: China’s Talent Recruitment Plans* (Washington, DC: U.S. Congress, November 2019), <https://www.hsgac.senate.gov/imo/media/doc/2019-11-18%20PSI%20Staff%20Report%20-%20China’s%20Talent%20Recruitment%20Plans%20Updated2.pdf>.
- 73 U.S. Senate Permanent Subcommittee on Investigations, *Threats to the U.S. Research Enterprise*.
- 74 “Clear Rules for Research Security and Researcher Responsibility,” The White House, August 10, 2021, <https://www.whitehouse.gov/ostp/news-updates/2021/08/10/clear-rules-for-research-security-and-researcher-responsibility/>.
- 75 For proposals and discussion, see e.g., Daniel Kliman et al., *Forging an Alliance Innovation Base* (Washington, DC: Center for a New American Security, March 2020), <https://www.cnas.org/publications/reports/forging-an-alliance-innovation-base>; Andrew Imbrie et al., *Agile Alliances* (Washington, DC: Center for Security and Emerging Technology, February 2020), <https://cset.georgetown.edu/publication/agile-alliances/>; and Zwetsloot, *China’s Approach to Tech Talent Competition*.
- 76 For a longer list of Chinese student destinations abroad, see Zwetsloot, *China’s Approach to Tech Talent Competition*, Table 1.

COVER PHOTO CHINA PHOTOS/GETTY IMAGES



1616 Rhode Island Avenue NW

Washington, DC 20036

202 887 0200 | www.csis.org